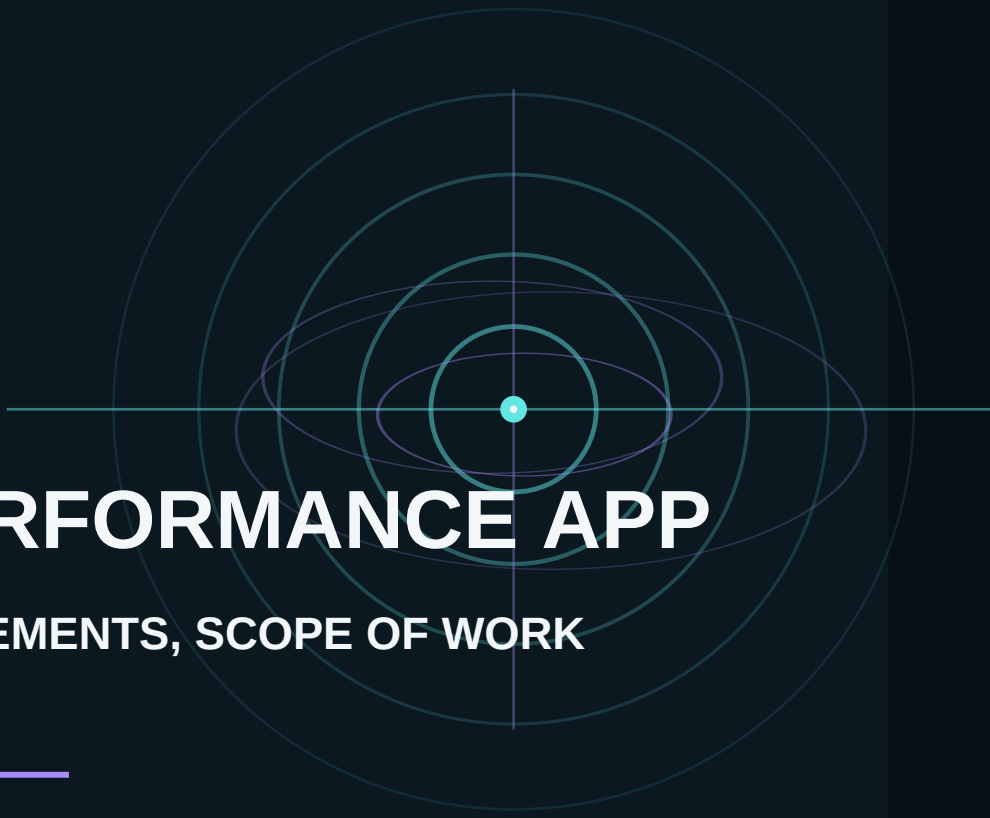
PSA-PRD-001 / VERSION 2.0



# PADEL PERFORMANCE APP

## PRODUCT REQUIREMENTS, SCOPE OF WORK & DESIGN BIBLE

Red-team reviewed development baseline

Sensor feasibility / Premium UX / Apple Intelligence / Production V1

iOS 27 | watchOS 27 | APPLE INTELLIGENCE

24 AUG 2026

PREMIUM APPLE-NATIVE PRODUCT + ENGINEERING GUIDANCE

## Document Control

| Field | Value |
|---|---|
| Document | Padel Performance App - Product Requirements, Scope of Work & Design Bible |
| Document ID | PSA-PRD-001 |
| Version | 2.0 - Red-Team Reviewed |
| Date | 24 Aug 2026 |
| Status | Complete development baseline - subject to owner approval and exact-state implementation gates |
| Internal prototype | PadelStrokeLab |
| Final product name | TBD |
| Target platforms | iOS 27+ and watchOS 27+ |
| Primary validation hardware | Apple Watch Ultra 2 on racket hand + compatible iPhone |
| Technical truth-check | 24 Aug 2026 |

> ***Design mandate:*** *Build a premium Apple-native padel performance instrument. The Watch measures; the scoring engine interprets; the iPhone reveals patterns; Apple Intelligence explains and coaches. Measurement authority never belongs to generative AI.*

## Contents at a Glance

| Part | Coverage |
|---|---|
| I - Product & evidence | Sections 1-18: vision, users, scientific basis, sensing, scoring, Core ML, Apple Intelligence role |
| II - Experience & premium design | Sections 19-25, 40-41, 57-58: iPhone/Watch UX, Liquid Glass, visual system, motion, haptics, copy, charts, screen catalogue |
| III - Data & architecture | Sections 26-34, 42-46, 59-61, 66-67: data model, sync authority, architecture, concurrency, repo blueprint, signal processing, performance budgets |
| IV - Intelligence | Sections 16-18, 63-64: Core ML, Foundation Models, PCC, AI Coach, Ask Coach, App Intents/Siri |
| V - Privacy & reliability | Sections 28-32, 47, 65: retention, HealthKit/privacy, failure handling, battery, accessibility, App Store, security/compliance |
| VI - Feasibility & validation | Sections 35-39, 62, 68: Lab Mode, GO/NO-GO, physical protocol, automated/physical/AI evidence |
| VII - Delivery & governance | Sections 49-55, 69-70: phases, |

| Part | Coverage |
|---|---|
| | Definition of Done, acceptance, open decisions, governance, A-to-Z delivery |
| VIII - Authority references | Sections 71-72: current Apple/research references and final product statement |

# 1. Purpose of this document

This document is the product, functional, technical, data, Artificial Intelligence (AI), user-experience, visual-design, testing and acceptance baseline for the full Padel Performance App.

It is intended to become the single product/design requirements bible used by product, design, engineering, Codex, testing and future reviewers.

The document defines **what the app is**, **what it is not**, **how it should behave**, **how it should look**, **what data is authoritative**, **how Apple Intelligence may be used**, **what must be proven physically**, and **what conditions must be met before a feature can be called ready**.

Engineering governance, Git process and active Gate Cards remain separate control documents. If there is a conflict:

1. Engineering governance controls process and gates.
2. This document controls approved product behavior and design intent.
3. The active Gate Card controls the current bounded implementation task.
4. Verified tests and physical evidence control claims about actual behavior.

No chat message, mock-up, AI output or assumption overrides the approved repository documents and verified code state.

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# 2. Product vision

## 2.1 Product idea

Create a premium Apple Watch + iPhone padel training application that uses the Apple Watch worn on the **racket-hand wrist** to measure the physical signature of racket strokes and provide objective, personalized feedback about **contact quality, repeatability and stroke execution**.

The product should answer a simple player question:

> ***"How clean and repeatable was my contact, and what should I work on next?"***

The app should work without requiring a sensor attached to the racket, a camera on court, a radar system or a professional coach.

The Apple Watch is the primary motion sensor. The iPhone is the primary analysis, history and coaching interface.

## 2.2 Core proposition

The product combines:

**Measure → Understand → Improve**

### Measure

- Detect racket swings and ball impacts.
- Capture high-rate wrist acceleration during impact.
- Capture wrist/device rotation and motion.
- Measure repeatable features around contact.

### Understand

- Estimate contact quality.
- Distinguish high-confidence clean contact from likely off-centre contact.
- Measure consistency relative to the same player.
- Classify stroke families where validated.
- Separate measured facts from inferred interpretation.

### Improve

- Show immediate, simple feedback on Apple Watch.
- Show detailed session analysis on iPhone.
- Compare the player with their own baseline and history.
- Use Apple Intelligence to explain results and create useful coaching summaries.
- Suggest practical next-session focus areas based only on verified data.

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## 3. Product principles

The following principles are mandatory and take precedence over convenience.

### P-001 - Measurement before intelligence

AI must never invent, modify or override sensor measurements, classifications, confidence values or scores.

### P-002 - Personal baseline before universal judgement

The app should primarily compare a player with their own validated patterns. Universal technique claims must be avoided unless supported by verified evidence.

### P-003 - Contact quality is not ball-placement accuracy

The Watch cannot determine where the ball lands. The product must not describe wrist-sensor output as court-placement accuracy.

### P-004 - Confidence is visible

When the system is uncertain, it must say so. A low-confidence stroke must not be presented as a certain conclusion.

### P-005 - Core functionality works without AI

Stroke detection, measurement, scoring, session storage and basic summaries must remain usable if Apple Intelligence, Private Cloud Compute or network connectivity is unavailable.

### P-006 - Premium means restrained, not overloaded

The product must feel like a precision sports instrument. It must not become a dashboard full of gauges, neon decoration or unnecessary science-fiction effects.

### P-007 - Watch is glanceable; iPhone is exploratory

The Watch provides immediate essential information. The iPhone provides detail, trends, comparison and coaching.

### P-008 - Raw evidence remains auditable

During development and beta, raw or minimally processed impact data must be retained sufficiently to reproduce and challenge scoring decisions.

### P-009 - No fake precision

Do not claim exact racket-head speed, exact sweet-spot coordinates, spin, ball speed or landing accuracy unless a future validated method proves them.

### P-010 - Graceful degradation

Unsupported sensor, missing permission, network loss, AI quota, sync delay or low-confidence data must produce a clear degraded state rather than fabricated results.

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## 4. Scientific and platform basis

### 4.1 Padel impact hypothesis

Published padel research has shown that impacts around a racket's sweet-spot region produce lower measured acceleration severity at the handle area than impacts farther from that region. A 2023 study evaluated 225 padel rackets and used the integral of squared acceleration during the first 40 milliseconds after contact as a vibration-severity measure.

This provides a credible physical basis for testing whether an Apple Watch on the racket wrist can distinguish cleaner contact from off-centre contact.

### 4.2 Wrist evidence from racket sports

Human tennis research has found substantially higher wrist/arm acceleration from off-centre impacts than centre impacts, with approximately threefold higher acceleration reported in one study. This supports the feasibility of measuring useful impact information at the wrist rather than only on the racket.

### 4.3 Apple Watch high-rate motion capability

Apple's `CMBatchedSensorManager` is explicitly intended for workout-centric use cases that benefit from higher-rate motion data without very tight latency requirements. Apple documents batches delivered roughly once per second, with up to **800 Hz accelerometer** and **200 Hz device-motion** samples, compared with up to 100 Hz through `CMMotionManager`. Apple also demonstrates impact detection and missed-swing separation in a racket/bat-style sports example.

This creates two requirements:

- The application shall detect actual runtime support (`isAccelerometerSupported` and the corresponding device-motion capability) rather than assume support from a marketing model name.
- Product design shall not promise sub-100-millisecond impact feedback from the batched stream. A realistic user-feedback target is one batch cycle plus processing, and this latency must be verified on physical hardware.

## 4.4 Workout and authorization requirement

High-rate sensing is workout-centric. The Watch app shall use an active HealthKit workout session while high-rate motion capture is active. The app shall also provide the required Motion usage description and HealthKit capabilities/usage descriptions before requesting authorization.

The implementation shall distinguish:

- Motion authorization.
- HealthKit authorization.
- Workout-session state.
- High-rate sensor capability.

A failure in any one of these shall prevent a false **Ready** state.

## 4.5 Apple Intelligence basis

On the 27 platform releases, Apple's Foundation Models framework provides a common Swift interface for Apple Foundation Models and other conforming language-model providers. For this product, the approved default hierarchy is:

1. Deterministic analytics and templates - always available.
2. `SystemLanguageModel` where available and validated - on-device, offline-capable, 4K context, no reasoning mode.
3. `PrivateCloudComputeLanguageModel` (PCC) only when the feature genuinely benefits from stronger reasoning or larger context - network required, 32K context, daily per-user quota, managed entitlement and developer eligibility required.

Apple makes PCC available on watchOS 27, but availability still depends on the device, region, Apple Intelligence state, entitlement, network and quota. The app must check runtime availability and must not make the core sports experience depend on PCC.

## 4.6 Signal-band limitation

An 800 Hz sampled accelerometer stream can only represent signal content below the Nyquist limit of 400 Hz. The published padel laboratory study sampled at 30 kHz and low-pass filtered at 1.5 kHz, so the Watch cannot reproduce the full laboratory vibration spectrum.

Therefore the product claim is deliberately narrower:

> ***We are testing whether the band-limited acceleration transmitted to the player's wrist contains enough repeatable information to estimate contact quality.***

The app shall never market itself as measuring the complete racket vibration spectrum.

## 4.7 Beta technology rule

Any feature relying on beta SDKs or beta entitlements shall remain feature-flagged until Apple ships the final SDK and the implementation is revalidated. No beta-only behavior may be treated as production evidence for App Store readiness.

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# 5. Target users

## 5.1 Primary user - recreational improvement player

A player who:

- Plays padel regularly.
- Wants objective improvement feedback.
- Already owns an Apple Watch.
- Does not want to attach hardware to the racket.
- Does not need professional biomechanical analysis.

Primary need:

> *"Tell me whether my contact is becoming cleaner and more consistent."*

## 5.2 Secondary user - performance-focused player

A player who wants:

- Stroke-by-stroke history.
- Trend analysis.
- Comparison of stroke families.
- Power/intensity versus contact-quality trade-off.
- Personal baseline tracking.

## 5.3 Future user - coach

A coach may later:

- Review a player's sessions.
- Compare training blocks.
- Add labels to selected strokes.
- Review clips/data exported by the player.

Coach features are not required for initial product validation.

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# 6. Hardware and operating assumptions

## 6.1 Mandatory assumption

**The Apple Watch must be worn on the racket-hand wrist during measurement.**

The app shall clearly communicate this during onboarding and before every first session after hand configuration changes.

## 6.2 Watch fit

The Watch must be reasonably snug and in a normal sports-wear position. A loose Watch can alter the measured signal.

The app shall provide simple fit guidance but must not pretend it can always detect incorrect strap tightness.

### 6.3 Racket

The product shall support normal padel rackets without an attached sensor.

Different rackets can produce different vibration behavior. The app shall therefore support a simple optional **Racket Profile** and shall not assume one universal vibration threshold works for every racket.

### 6.4 Platform baseline

Initial full-product architecture targets:

- iOS 27 or later for the full premium/Apple Intelligence experience.
- watchOS 27 or later for the full premium/Apple Intelligence Watch experience.
- SwiftUI and the current Apple design system.
- HealthKit workout session for high-rate sensing.
- Core Motion, including `CMBatchedSensorManager` when supported.
- WatchConnectivity for app-specific synchronization unless a better Apple-native mechanism is explicitly approved for a bounded use case.
- Foundation Models framework for Apple Intelligence features.
- App Intents for Siri/Spotlight/Shortcuts/system actions.
- Core ML for future on-device stroke classification only after physical data validation.
- Swift Charts for iPhone analytics where appropriate.
- OSLog/Unified Logging with privacy-safe metadata.

The feasibility logger shall use only the subset necessary to prove sensing and shall remain structurally separate from consumer features.

---

## 7. Product boundaries - what the app can and cannot claim

### 7.1 Intended measurable/inferable outputs

Subject to physical validation, the product may provide:

- Stroke count.
- Impact detection.
- Air swing versus ball-contact classification.
- Contact-quality score.
- Likely clean versus likely off-centre contact.
- Post-impact vibration/acceleration characteristics.
- Wrist rotational characteristics.
- Stroke smoothness proxies.
- Follow-through consistency proxies.
- Relative stroke intensity.
- Personal consistency.
- Stroke-family classification.
- Session and historical trends.

### 7.2 Outputs that are prohibited without future evidence

The application must not claim from Watch-only sensing:

- Exact ball landing position.
- Court-placement accuracy.
- Exact racket sweet-spot coordinate.
- Exact ball speed.
- Exact racket-head speed.
- Exact ball spin.
- Tactical effectiveness of the shot.
- Whether the shot won a point.
- Medical injury diagnosis or injury prevention guarantee.

If future sensors or validated techniques enable these, this document must be revised before implementation.

---

# 8. End-to-end user journey

## 8.1 First launch on iPhone

1. Premium welcome animation.
2. Explain the concept in one sentence.
3. Confirm racket hand: Left / Right.
4. Explain that Watch must be on racket hand.
5. Request required permissions progressively, not all at once.
6. Connect/verify Watch app.
7. Optional racket profile.
8. Guided personal calibration.
9. Show a short explanation of Contact Quality and Confidence.
10. Ready screen.

The entire onboarding should feel like configuring a precision sports instrument, not completing a form.

## 8.2 Starting a session

The player can start from:

- Apple Watch app.
- iPhone app.
- App Intent / Siri where supported.
- Future Action Button/Control integration if technically appropriate and approved.

The Watch must confirm:

- Sensor capability available.
- Workout started.
- High-rate capture running.
- Storage available.

The user must not see "Ready" until these are verified.

## 8.3 During a session

The Watch displays only essential information:

- Session active state.

- Shot count.
- Most recent high-confidence Contact Quality score.
- Session average or clean-contact rate.
- Optional subtle haptic feedback.

Detailed graphs remain on iPhone.

### 8.4 After a session

Watch:

- Overall session score.
- Clean-contact rate.
- One strongest area.
- One focus area.
- Optional short AI summary if available.

Phone:

- Full session dashboard.
- Stroke distribution.
- Contact-quality trend.
- Stroke-family breakdown.
- Intensity versus contact-quality relationship.
- Personal-baseline comparison.
- AI Coach summary.
- Ask Coach entry.

---

## 9. Functional requirements - onboarding and setup

### ONB-001 - Racket hand

The app shall require the user to choose left or right racket hand.

### ONB-002 - Watch position reminder

The app shall clearly state that Watch must be worn on the racket wrist.

### ONB-003 - Fit guidance

The app shall show a short visual instruction for normal snug sports fit.

### ONB-004 - Permissions

Permissions shall be requested at the moment their benefit is clear.

Required or potentially required permissions include:

- HealthKit workout access.
- Motion access where applicable.
- Notifications only if the user enables coaching reminders.

### ONB-005 - Racket profile

User can optionally define:

- Racket nickname.
- Brand/model text.
- Shape if desired.
- Date added.

No performance claim shall be inferred solely from racket marketing characteristics.

### ONB-006 - Calibration

The app shall provide a guided personal calibration after feasibility has proven the required signal quality.

Calibration shall be versioned and repeatable.

---

## 10. Functional requirements - session control

### SES-001 - Start

Starting a session shall create a unique session identifier before sensor capture begins.

### SES-002 - Active verification

A session is "Active" only after the workout and sensor pipelines confirm they are running.

### SES-003 - Pause

Pause shall stop classification/scoring according to the approved sensing behavior while preserving session state.

### SES-004 - Resume

Resume shall not create duplicate sessions or duplicate strokes.

### SES-005 - End

Ending a session shall flush pending data, save the session summary and close the workout cleanly.

### SES-006 - Recovery

If the app is interrupted or terminated unexpectedly, recoverable session data shall be preserved whenever platform rules allow.

### SES-007 - No phantom readiness

The UI shall never display a successful capture state if the sensor stream is unavailable.

---

# 11. Sensor acquisition requirements

## MOT-001 - High-rate accelerometer

Use `CMBatchedSensorManager` high-rate accelerometer when supported.

Target high-rate capability: **800 Hz**.

The runtime shall record the actual configured/received rate as evidence.

## MOT-002 - Device motion

Use high-rate device motion when validated as useful.

Target capability: **200 Hz**.

Device-motion features may include:

- Rotation rate.
- Gravity-relative orientation.
- Angular velocity.
- Pre-impact motion shape.
- Follow-through characteristics.

## MOT-003 - Monotonic timing

Every sample and detected stroke shall use a reliable monotonic timing basis suitable for aligning acceleration and device-motion streams.

## MOT-004 - Ring buffer

During an active session, the app should maintain a rolling in-memory buffer so only useful pre-impact/post-impact windows need long-term persistence.

## MOT-005 - Raw evidence

Development and beta builds must support preservation/export of raw impact windows.

## MOT-006 - No silent sample loss

Detected gaps, unsupported frequency, sensor errors or dropped batches shall be logged and surfaced in diagnostic evidence.

## MOT-007 - Battery-aware processing

The Watch shall not redraw the user interface at sensor frequency. Signal processing must be decoupled from UI updates.

## MOT-008 - Capability check

The app shall use actual Core Motion capability APIs and shall not rely only on Watch model names.

---

# 12. Stroke event detection

## DET-001 - Candidate swing detection

The system shall identify candidate racket-swing events from motion data.

## DET-002 - Impact detection

The system shall identify a likely ball-contact timestamp using the high-frequency acceleration signature.

## DET-003 - Air-swing rejection

Where confidence is sufficient, the system shall distinguish a swing without impact from a swing with impact.

## DET-004 - Debounce

One physical impact shall produce one stroke event.

## DET-005 - Noise rejection

Routine arm motion, walking, picking up a ball and similar movements shall not be counted as strokes when the classifier can confidently reject them.

## DET-006 - Uncertain events

Ambiguous events shall be marked uncertain rather than forced into a stroke class.

---

# 13. Contact-quality analysis

## 13.1 Initial physical feature

The initial feasibility feature shall include acceleration-energy analysis around contact, inspired by padel vibration research.

A provisional feature is:

**V40 = integral of acceleration magnitude squared over approximately the first 40 ms after detected contact.**

This is a research feature, not a guaranteed final scoring formula.

## 13.2 Candidate production features

Subject to validation, the scoring engine may use:

- Peak resultant acceleration.
- V40 acceleration energy.
- Post-impact energy over additional short windows.
- Ring-down/decay behavior.
- Ratios of post-impact energy to total impact intensity.
- Frequency-domain features within the Watch-observable range.
- Wrist angular velocity around impact.

- Pre-impact swing smoothness.
- Change in angular motion after contact.
- Follow-through repeatability.
- Player-specific normalization.
- Racket-profile normalization.

### SCR-001 - Contact Quality Score

The app shall provide a **Contact Quality Score (CQS)** on a 0-100 scale only after the score has been physically validated.

The score must represent quality of measured contact characteristics - not ball-placement accuracy.

### SCR-002 - Confidence

Every automatically classified stroke shall carry a confidence value.

### SCR-003 - Confidence threshold

Low-confidence strokes shall not be used as equally authoritative inputs to headline session metrics.

### SCR-004 - Personal baseline

Scores should be normalized primarily against the user's own validated baseline, appropriate stroke family and intensity range.

### SCR-005 - Algorithm versioning

Every scored stroke shall retain the algorithm/model version that generated its score.

### SCR-006 - Reproducibility

For retained raw evidence, a scoring version should be reproducible from the stored input and model/parameter version.

---

## 14. Session metrics

The product shall progressively support the following metrics after validation.

### 14.1 Headline metrics

- Stroke count.
- Contact Quality Score.
- Clean Contact Rate.
- Consistency Score.
- Relative Intensity.
- High-confidence stroke percentage.

### 14.2 Advanced metrics

- Forehand Contact Quality.
- Backhand Contact Quality.

- Volley Contact Quality.
- Overhead Contact Quality.
- Serve Contact Quality.
- Contact Quality by intensity band.
- Contact-quality trend during session.
- Best clean-contact streak.
- Variability/dispersion of contact-quality features.

### 14.3 Session score

A headline Session Score may be offered only if its composition is transparent and stable.

It should favor confidence-weighted robust statistics rather than a naive arithmetic mean that can be distorted by outliers.

---

## 15. Stroke-family classification

Stroke recognition is a later validated feature, not a dependency for feasibility.

Candidate classes:

- Forehand.
- Backhand.
- Forehand volley.
- Backhand volley.
- Serve.
- Overhead family.

More specific overhead classes such as smash, bandeja and vibora may be introduced only if physical data shows reliable separation.

The app shall not display a precise stroke label when confidence is below the approved threshold.

---

## 16. Core ML strategy

### 16.1 Purpose

Core ML may be used after the feasibility gate to improve classification of:

- Impact versus air swing.
- Clean versus poor/off-centre contact.
- Stroke family.
- Player-specific contact patterns.

### 16.2 Model philosophy

Start with simple, explainable models where possible. Complexity is justified only when it produces material validated improvement.

### 16.3 On-device inference

Production inference should run on-device wherever practical for latency, privacy and reliability.

### 16.4 Model versioning

Every prediction shall be associated with the exact model version.

### 16.5 Personalization

A global model may provide the initial prior, but user-specific calibration should refine thresholds/features where evidence supports it.

### 16.6 No AI rescue of weak physics

If physical testing shows insufficient separation between clean and poor contact, adding a more complex model is not automatically acceptable. The signal must contain repeatable information first.

---

## 17. Apple Intelligence - product role

Apple Intelligence is a **coaching and explanation layer**, not the measurement authority.

### AI-001 - Session Coach

After a session, Apple Intelligence may transform verified structured metrics into a concise coaching summary.

Example intent:

> *"Your forehand contact was more consistent today. Contact quality dropped when your relative swing intensity increased. Next session, keep the same contact pattern as you add power."*

The model must receive the underlying measured values and confidence information.

### AI-002 - Explain a result

The user can ask:

- Why did my score drop?
- What improved today?
- Which stroke is least consistent?
- What happens when I swing harder?

The model shall answer through approved data tools rather than guessing from conversational context.

### AI-003 - Ask Coach

The iPhone app shall provide a conversational **Ask Coach** experience after the measurement foundation is validated.

Candidate tools available to the model:

- `getSessionSummary(sessionID)`
- `compareSessions(sessionA, sessionB)`
- `getStrokeStats(type, range)`

- `getContactQualityByIntensity(range)`
- `getPersonalBaseline(strokeType)`
- `getLowestConfidenceStrokes(sessionID)`
- `getTrend(metric, period)`

## AI-004 - Structured output

Where the AI result drives UI components, use Foundation Models structured/guided generation rather than parsing free-form text.

## AI-005 - Model selection

Use the simplest available model that meets the quality requirement:

1. Deterministic template summary if AI unavailable.
2. On-device Apple model where appropriate and available.
3. Private Cloud Compute only for features that require stronger reasoning/context and where entitlement/network/quota permit.

## AI-006 - Watch AI

On watchOS 27, short glanceable coaching summaries may use the Foundation Models framework/PCC when available.

The Watch shall never block session completion waiting for AI.

## AI-007 - AI availability

If Apple Intelligence is unavailable, disabled, unsupported, offline or over PCC quota, the app shall provide a useful deterministic fallback without erroring the core experience.

## AI-008 - No metric mutation

AI output cannot change scores, labels, raw measurements or model confidence.

## AI-009 - Evidence-linked coaching

A coaching statement should be traceable to the session metrics used to generate it.

## AI-010 - No medical advice

AI must not diagnose injury or make unsupported medical claims. It may suggest reducing intensity or consulting a qualified professional only in clearly appropriate non-diagnostic language.

## AI-011 - No invented biomechanics

The model shall not prescribe precise technique unless the recommendation is supported by approved domain guidance and relevant measured data.

## AI-012 - Evaluation

AI features must have an evaluation suite covering factual grounding, unsupported claims, metric consistency, tool use, fallback behavior and concise tone.

---

# 18. Apple Intelligence user value hierarchy

The value of Apple Intelligence should be prioritized in this order:

### 1. Explain numbers

Turn sensor metrics into understandable language.

### 2. Find the important pattern

Surface one or two meaningful changes instead of forcing the user to interpret many charts.

### 3. Compare the user with themselves

Explain improvement or decline relative to personal history.

### 4. Give a next action

Recommend a simple, measurable focus for the next session.

### 5. Answer natural questions

Allow the player to query their own training data conversationally.

### 6. Keep Watch feedback glanceable

Use AI for short summaries, not paragraphs on the wrist.

---

# 19. iPhone information architecture

The iPhone application should use a maximum of four primary navigation areas.

## 19.1 Today

Purpose: immediate status and next action.

Contents:

- Latest session hero card.
- Current Contact Quality trend.
- Clean Contact Rate.
- One AI Coach insight.
- Start Session action.

## 19.2 Sessions

Purpose: historical exploration.

Contents:

- Session list.
- Session detail.
- Compare sessions.
- Stroke explorer.

- Trend charts.

### 19.3 Coach

Purpose: interpretation and improvement.

Contents:

- AI Session Coach.
- Ask Coach.
- Current focus.
- Suggested drill/focus.
- Improvement narrative.

### 19.4 Profile / Setup

Contents:

- Racket hand.
- Racket profiles.
- Calibration.
- Watch status.
- Data/export.
- AI/privacy settings.
- Notifications.
- About/scoring explanation.

---

## 20. Apple Watch information architecture

Watch interactions must be shallow and glanceable.

### 20.1 Ready screen

- Product mark/name.
- Watch sensor ready indicator.
- Large Start button.

### 20.2 Active session

Primary screen:

- Session active indicator.
- Large last Contact Quality score.
- Stroke count.
- Clean Contact Rate or session average.

Secondary page via Digital Crown/swipe:

- Forehand/backhand breakdown when validated.
- Session time.
- Sensor/connection diagnostic only if something requires attention.

### 20.3 Pause/end

Use familiar, large controls. Avoid small multi-button toolbars.

### 20.4 Post-session

- Session score.
- Clean Contact Rate.
- Strongest pattern.
- One focus point.
- Sync state.

### 20.5 Always On

When Always On is active during a frontmost workout experience:

- Reduce animation.
- Dim nonessential content.
- Preserve essential session state and score/status.
- Avoid exposing unnecessarily sensitive detail.

---

## 21. Premium design direction

### 21.1 Design concept

**Precision Sports Instrument + Liquid Glass**

The app should feel like a future-facing professional instrument built by Apple-platform specialists, not a generic fitness template.

Keywords:

- Precise.
- Calm.
- Premium.
- Technical.
- Fast.
- Tactile.
- Personal.
- Intelligent.

Avoid:

- Cartoon sports graphics.
- Fake chrome.
- Excessive neon.
- Gaming-style badges everywhere.
- Dense dashboards.
- Unnecessary 3D decoration.
- Glass applied to every card.

### 21.2 iOS 27 / watchOS 27 native-first design

Use native SwiftUI components and current Apple platform conventions so the app naturally adopts current Liquid Glass behavior.

Liquid Glass is a functional layer for controls/navigation. It must not be used indiscriminately across the content layer.

Use:

- Native tab/navigation structures.
- Native materials.
- Edge-to-edge content.
- System spacing and interaction behavior.
- SF Symbols for interface iconography.
- Native accessibility behavior.

### 21.3 Visual hierarchy

Hierarchy should be:

1. **Primary metric** - large and unmistakable.
2. **Interpretation** - one short phrase.
3. **Context** - comparison/baseline/confidence.
4. **Detail** - charts and raw metrics on demand.

The user should understand the screen in under one second on Watch and within several seconds on iPhone.

---

## 22. Design system

### 22.1 Color philosophy

Dark-first, adaptive, high-contrast premium identity.

Provisional semantic palette:

- **Obsidian** - deep neutral content background.
- **Polar** - primary text/highlight.
- **Ion** - cool cyan/teal performance accent.
- **Violet** - intelligence/coach accent.
- **Amber** - caution/uncertain.
- **Coral** - poor/attention state.

Exact colors are design tokens and must be tested for accessibility, Light/Dark appearance and wide-gamut displays.

Do not encode status by color alone.

### 22.2 Liquid Glass

Use Liquid Glass for:

- Primary controls.
- Tab/navigation layer.

- Floating actions.
- Transient interactive overlays.

Do not use Liquid Glass for:

- Every data card.
- Full-screen content backgrounds.
- Dense chart surfaces.

### 22.3 Typography

Use Apple system typography.

Principles:

- SF system families only unless a future approved brand requirement justifies otherwise.
- Large scores use monospaced digits where stable alignment improves readability.
- No tiny technical labels on Watch.
- Dynamic Type supported on iPhone.
- Numerical hierarchy should feel instrument-grade.

### 22.4 Shapes

Use native corner behavior and component geometry.

Avoid a custom "rounded rectangle everywhere" design language that conflicts with current Apple controls.

### 22.5 Iconography

Use SF Symbols for interface actions.

Do not use SF Symbols as the brand logo or app icon.

App icon shall be a custom layered design, simple and recognizable at small size.

### 22.6 App icon concept

Direction:

- Abstract racket-impact/contact wave.
- Strong central geometry.
- Layered depth suitable for current Apple icon system.
- Premium sports/measurement feel.
- No text.
- Avoid pure black Watch icon background that visually disappears into the screen.

Final icon requires separate design approval.

---

## 23. Motion and interaction design

Motion must communicate state, not decorate it.

### 23.1 Impact response

When a new high-confidence stroke arrives:

- Score transitions smoothly.
- A short controlled radial/energy response may appear.
- Clean contact should feel precise and settled.
- Poor/uncertain contact may show a less stable but still restrained transition.

Do not attempt to visually simulate physical vibration literally.

### 23.2 Score reveal

Target feel:

- Fast.
- Confident.
- Under approximately one second.
- No blocking animation.

### 23.3 Reduced Motion

All custom motion must respect Reduce Motion and relevant accessibility settings.

### 23.4 Haptics

Haptics should be optional and purposeful.

Proposed modes:

- Off.
- Important only.
- Coach mode.

Default should not produce a strong haptic after every normal shot.

Possible haptic uses:

- Session starts/stops.
- High-confidence poor-contact pattern if Coach mode enabled.
- Personal best or clean streak milestone.
- Critical sensor/session error.

---

## 24. Copy and tone

Product language shall be:

- Clear.
- Short.
- Precise.
- Non-judgmental.
- Evidence-based.

Prefer:

- "Clean contact"
- "Likely off-centre"
- "More consistent than your baseline"
- "Low confidence"
- "Not enough data yet"

Avoid:

- "Terrible shot"
- "Wrong technique"
- "Perfect stroke" unless the product defines exactly what perfect means.
- Unsupported injury or coaching claims.

AI coaching should explain **why** it is making a statement with specific metrics when useful.

---

# 25. Data visualization

## 25.1 iPhone charts

Preferred charts:

- Contact Quality over time.
- Distribution of Contact Quality.
- Contact Quality versus relative intensity.
- Stroke-family comparison.
- Weekly/monthly trend.

Keep charts simple and interpretable.

## 25.2 Watch charts

Avoid complex interactive charts on Watch.

Use:

- One sparkline.
- One simple ring/progress visualization.
- One compact trend arrow.

Detailed exploration belongs on iPhone.

## 25.3 Accessibility

Every chart shall have:

- Clear title.
- Units/meaning.
- Accessibility description.
- Non-color-only differentiation.

---

# 26. Data model

## 26.1 Session

Minimum fields:

- `sessionID`
- `startTime`
- `endTime`
- `sourceWatchID` or anonymous device fingerprint suitable for diagnostics
- `appBuild`
- `watchOSVersion`
- `sensorCapability`
- `actualAccelerometerRate`
- `actualDeviceMotionRate`
- `algorithmVersion`
- `modelVersion`
- `calibrationVersion`
- `racketProfileID`
- `racketHand`
- `strokeCount`
- `qualitySummary`
- `syncState`

## 26.2 Stroke

Minimum fields:

- `strokeID`
- `sessionID`
- `impactTimestamp`
- `strokeType`
- `strokeTypeConfidence`
- `contactQualityScore`
- `contactConfidence`
- `impactPeak`
- `V40`
- Additional approved impact features.
- Rotation/motion features where used.
- Relative intensity.
- User label/override if research mode permits.
- `algorithmVersion`
- `modelVersion`
- Raw snippet reference.

## 26.3 Calibration profile

- Player/racket context.
- Creation date.
- Calibration protocol version.
- Feature baselines.

- Stroke-family baselines.
- Intensity bands.
- Validity status.

### 26.4 AI insight

- `insightID`
- Source session/range.
- Model path: deterministic/on-device/PCC.
- Prompt/instruction version.
- Tool calls/metric references needed for audit.
- Generated text/structured result.
- Generation timestamp.

AI insight data shall never be the source of truth for measured metrics.

---

## 27. Watch-iPhone authority and synchronization

### 27.1 Live authority

During a session, the Watch is authoritative for:

- Raw motion capture.
- Impact event timestamps.
- Locally generated stroke IDs.
- Locally persisted unsynced session data.

### 27.2 Long-term authority

After verified synchronization, the iPhone app is authoritative for:

- Long-term session library.
- Historical analysis.
- AI coaching context.
- User-facing trend history.

### 27.3 Sync rules

- Every object uses stable unique IDs.
- Sync must be idempotent.
- Reconnect must not duplicate strokes.
- Watch data is retained until transfer is confirmed according to retention policy.
- "Synced" may only appear after confirmed persistence on the receiving side.
- Conflicts must be deterministic and logged.

### 27.4 Offline operation

A user must be able to complete a Watch session without the iPhone nearby where platform capability permits.

AI coaching can be delayed.

---

# 28. Storage and retention

## 28.1 Production principle

Do not store an entire continuous 800 Hz session indefinitely unless needed.

Preferred production strategy:

- Keep compact derived stroke features long-term.
- Keep short raw windows around impacts for a defined period or user-selected research mode.
- Make full raw recording a development/beta/research option.

## 28.2 Development and beta

During feasibility and algorithm validation:

- Preserve raw data needed to reproduce results.
- Support export.
- Record app/OS/device/model versions.

## 28.3 Export

Development/beta export should support:

- CSV for sensor/features.
- JSON for structured session metadata.

A polished user-facing PDF training report is a later optional feature.

---

# 29. Privacy and security

## PRIV-001 - Local first

Sensitive motion, workout and personal performance data should remain on-device by default wherever practical.

## PRIV-002 - Permission minimization

Request only permissions necessary for enabled features.

## PRIV-003 - HealthKit

HealthKit data access follows Apple's privacy and purpose restrictions.

## PRIV-004 - Apple Intelligence

When using Apple on-device models, data stays within the supported Apple Intelligence model path.

When using PCC, the feature must follow Apple's entitlement, availability and privacy model.

## PRIV-005 - External AI

No third-party model receives user performance/health data unless a future approved feature introduces it with explicit architecture, disclosure and consent.

### PRIV-006 - No hidden analytics

Any analytics/telemetry must be documented and minimized.

### PRIV-007 - Deletion

User shall be able to delete sessions and their associated app-owned analysis data.

### PRIV-008 - Export

User shall be able to export their app-owned performance data in a reasonable format in a later production stage.

---

## 30. Reliability and failure behavior

### FAIL-001 - Unsupported Watch sensor

Do not generate a fake high-rate score. Explain that the required sensor mode is unavailable.

### FAIL-002 - Health permission denied

Explain what is required and how the feature is affected. Do not claim active measurement.

### FAIL-003 - Workout fails to start

Session remains Not Ready.

### FAIL-004 - Sensor batch error

Log error, stop authoritative scoring if necessary, and clearly mark affected time/data.

### FAIL-005 - Sample-rate degradation

Record actual behavior and mark affected session evidence.

### FAIL-006 - Watch-iPhone disconnect

Continue safely on Watch where possible, queue data and sync later.

### FAIL-007 - Low battery

Warn when reasonable; do not silently reduce measurement quality without recording it.

### FAIL-008 - AI unavailable

Use deterministic summary or display measured metrics without AI.

### FAIL-009 - PCC quota/network issue

Fall back gracefully; session data and scoring remain unaffected.

### FAIL-010 - Low classification confidence

Display "Uncertain" or exclude from headline metrics according to approved logic.

### FAIL-011 - Interrupted session

Persist recovery checkpoints where allowed.

### FAIL-012 - Corrupt/partial data

Never infer missing samples as measured values.

---

## 31. Performance and battery requirements

The product must be useful for real padel sessions without unacceptable battery drain.

Requirements:

- Batch processing rather than per-sample UI work.
- Short raw windows retained around candidate events.
- Efficient ring buffers.
- No AI inference during every impact unless later proven necessary and battery-safe.
- AI coaching should primarily happen after sessions.
- Background work follows platform rules.
- Performance instrumentation included during beta.

No battery-life claim shall be made until physical sessions are measured on target hardware.

---

## 32. Accessibility requirements

The app shall support:

- Dynamic Type on iPhone where applicable.
- VoiceOver labels.
- Sufficient contrast.
- Reduce Motion.
- Reduce Transparency where relevant.
- Non-color-only state communication.
- Large, reachable Watch controls.

Accessibility is part of premium quality and not a post-launch enhancement.

---

## 33. Notifications, widgets and complications

These are secondary to the core experience.

### 33.1 Notifications

Optional uses:

- Session processing complete.
- Weekly progress summary.
- User-configured training reminder.

Avoid noisy motivational notifications by default.

### 33.2 Watch complication / Smart Stack

Potential content:

- Last session Contact Quality.
- Weekly clean-contact trend.
- Start session shortcut.

### 33.3 iPhone widgets

Potential content:

- Last session.
- Weekly trend.
- Current focus.

Widgets must remain useful without revealing sensitive data a user might not want on the Lock Screen; privacy settings must be respected.

---

## 34. Siri and App Intents

Future approved intents may include:

- Start padel session.
- End padel session.
- Show last session.
- Compare my last two sessions.
- What should I work on today?

App Intents shall call authoritative app functions; Siri/AI text is not the data authority.

---

## 35. Lab / research mode

The product codebase should preserve a controlled internal **Lab Mode** for development and beta validation.

Lab Mode may expose:

- Actual sensor rate.
- Batch counts.
- Dropped-sample indicators.
- Raw waveform.

- Manual trial labels.
- Centre/off-centre test labels.
- Air-swing labels.
- Export controls.
- Algorithm/model version.

Lab Mode must not clutter the consumer interface.

---

## 36. Feasibility gate - GO / NO-GO

The first product gate remains intentionally simple.

### Test A - Sensor acquisition

PASS only if the target Watch reliably provides the expected high-rate stream during a valid workout and data can be persisted correctly.

### Test B - Air swing versus impact

Perform controlled air swings and ball contacts.

PASS if impacts are consistently and clearly identifiable and false impacts are not generated by the air swings in the agreed test set.

### Test C - Controlled centre versus off-centre racket impacts

Use repeated, labelled centre and deliberately off-centre impacts with the same racket and ball.

Initial threshold:

- At least 80% correct separation/classification in the agreed controlled set.

### Test D - Real self-fed/simple strokes

Use deliberately clean and deliberately poor/off-centre real strokes.

Initial threshold:

- At least 80% correct classification in the agreed test set.

### Decision

**GO** only if the physical signal supports repeatable separation and there is no evidence-integrity problem.

**NO-GO** if the Watch-only signal does not reliably separate the classes after technical capture issues have been ruled out.

Do not add sophisticated machine learning merely to force a GO result.

---

## 37. Product validation after feasibility

A feasibility GO is not a production-validation result.

Before commercial claims, broaden testing across:

- Multiple players.
- Left and right racket hands.
- Different skill levels.
- Different Watch bands/fit.
- Different racket models.
- Different shot intensities.
- Forehand/backhand/volley/overhead families.
- Real match distractions and non-stroke arm motion.

The purpose is to establish robustness and identify where personalized calibration is mandatory.

---

# 38. Testing requirements

## 38.1 Unit tests

Cover:

- Signal-window calculations.
- Timestamp alignment.
- V40 calculation.
- Feature normalization.
- Session aggregation.
- Confidence weighting.
- Data serialization.
- Sync deduplication.
- AI structured-output parsing.

## 38.2 Integration tests

Cover:

- Workout lifecycle.
- Sensor lifecycle.
- Session persistence.
- Watch-iPhone sync.
- Offline queue/reconnect.
- AI fallback behavior.

## 38.3 Regression tests

Every confirmed bug affecting scoring, persistence or sync gets a regression test where practical.

## 38.4 Simulator limits

Simulators cannot prove real high-rate wrist impacts, physical Watch behavior, haptics, battery or connectivity behavior.

### 38.5 Physical device tests

Relevant production changes require physical Apple Watch + iPhone evidence.

### 38.6 AI evaluations

Evaluate:

- Grounding in actual metrics.
- No invented session facts.
- Tool-call correctness.
- Conciseness.
- No score mutation.
- Safe fallback.
- No medical diagnosis.

---

## 39. Quality targets

No marketing promise is approved until validated, but engineering targets should include:

- No silent data corruption.
- No duplicate stroke IDs after reconnect.
- No session marked synced until confirmed.
- No score when sensor evidence is insufficient.
- No AI statement presented as a measurement.
- Fast Watch interaction with no noticeable UI stalls from sensor processing.
- Graceful degraded state without network/AI.
- Repeatable algorithm output for the same retained input/version.

Classification accuracy targets must be defined per validated feature and dataset rather than one universal number.

---

## 40. Premium UI screen requirements

### 40.1 iPhone - Today

Hero area:

- Large last-session score.
- Short interpretation.
- Trend versus personal baseline.
- Start Session glass control.

Secondary:

- Clean Contact Rate.
- Consistency.
- Current focus.
- AI Coach insight card.

### 40.2 iPhone - Session detail

Top:

- Session score.
- Date/duration.
- Confidence coverage.

Middle:

- Contact Quality chart.
- Stroke-family cards.
- Intensity versus quality.

Bottom:

- AI Coach.
- Stroke Explorer.
- Compare.
- Export where enabled.

### 40.3 iPhone - Stroke detail

- Contact Quality.
- Confidence.
- Stroke family.
- Relative intensity.
- Simple impact waveform in Lab/advanced mode.
- Explain This Stroke action.

### 40.4 iPhone - Coach

- Current training focus.
- Improvement narrative.
- Ask Coach conversational entry.
- Source metrics available on demand.

### 40.5 Watch - Active

- Very large last score.
- Compact label: Clean / Likely off-centre / Uncertain.
- Stroke count.
- Session trend.

No dense charts.

---

## 41. Premium interaction examples

### Example A - clean stroke

Watch receives high-confidence clean contact.

UI:

- Score changes to 88.
- Brief precise radial response.
- Label: **Clean**.
- No strong haptic by default.

### Example B - likely poor contact

UI:

- Score 54.
- Label: **Likely off-centre**.
- Optional Coach Mode haptic.

### Example C - uncertain

UI:

- No misleading numeric judgement if confidence is below threshold.
- Label: **Uncertain**.
- Stroke can still count if impact is certain but quality is not.

### Example D - post-session AI

> *"Your clean-contact rate improved from your recent baseline. The improvement came mainly from medium-intensity forehands. At higher intensity your contact became less consistent."*

This text is permitted only when the referenced metrics support it.

---

## 42. Technical architecture

Recommended logical layers:

```
Watch UI
   ↓
WorkoutSessionController
   ↓
MotionCaptureService
   ↓
StrokeEventDetector
   ↓
FeatureExtractor
   ↓
ContactScoringEngine / CoreMLClassifier
   ↓
Local Session Store
   ↓
Sync Queue
   ↓
_____________________________
        iPhone
_____________________________
Sync Receiver / Repository
   ↓
Session Analytics
   ↓
Trend Engine
   ↓
Apple Intelligence Tools
```

```
   ↓
AI Coach / Ask Coach
   ↓
SwiftUI Presentation
```

Cross-cutting services:

- Versioning.
- Diagnostics.
- Privacy.
- Export.
- Feature flags.
- Accessibility.

Production code should maintain clear boundaries so AI, UI or sync changes cannot silently change the physics/scoring layer.

---

## 43. Concurrency and data-integrity requirements

High-rate motion capture, workout lifecycle, persistence and sync require explicit concurrency design.

Requirements:

- Single authoritative session state machine.
- Ordered stroke event creation.
- Stable IDs before sync.
- Thread-safe sensor buffers.
- No UI thread high-rate processing.
- Cancellation-safe stop/end behavior.
- Idempotent persistence.
- Idempotent sync.
- Clear ownership of mutable session state.

Concurrency shortcuts that introduce duplicate, reordered or lost strokes are release blockers.

---

## 44. Algorithm transparency

The app shall include a user-facing **How scoring works** explanation that is understandable without engineering knowledge.

It should state:

- Watch measures motion at the racket wrist.
- App looks at the impact signature and repeatability.
- Score compares the stroke with validated patterns and personal baseline.
- Score does not know where the ball landed.
- Confidence reflects how certain the system is.

Advanced technical details may be available separately.

---

## 45. Versioning and change control

Changes that alter the meaning of a score require:

- New algorithm/model version.
- Updated tests.
- Re-evaluation of affected physical evidence.
- Migration strategy for historical comparisons where necessary.

The app must not silently compare scores produced by materially incompatible scoring versions as if they were identical.

---

## 46. Feature flags

Use explicit feature flags for incomplete or experimental features such as:

- Core ML classifier.
- AI Coach.
- PCC reasoning.
- Stroke-family recognition.
- Advanced charts.
- Racket normalization.
- Lab Mode.

A hidden feature must not accidentally affect production scoring.

---

## 47. App Store and launch requirements

Before public release:

- Use a production Xcode/SDK accepted by Apple.
- Final privacy policy, App Privacy disclosures and privacy manifest are complete and consistent with runtime behavior.
- HealthKit usage descriptions are accurate, minimal and understandable.
- Health/fitness data is not used for advertising, behavioral targeting or unrelated data mining.
- HealthKit-sourced data is not written inaccurately and is not copied to iCloud merely for convenience; any future cloud design requires a specific data-classification/privacy review.
- AI/PCC entitlement and developer-eligibility status are resolved if PCC is shipped.
- AI availability/quota states are reviewable without blocking the app.
- Unsupported-device behavior is tested.
- Accessibility review is complete.
- Physical Watch+iPhone test matrix is complete.
- App Store screenshots and marketing do not overstate measurement capability.
- Review Notes explain any hardware-dependent workflow and provide a demo path/video if App Review cannot reasonably reproduce the racket workflow.
- Terms must not describe the app as a medical device unless future regulatory work explicitly changes scope.

---

# 48. Non-goals for first commercial version

Unless later approved, V1 does not require:

- Video analysis.
- Court cameras.
- Radar integration.
- External racket sensors.
- Social network/feed.
- Match scoring system.
- Player matchmaking.
- Tournament management.
- Coach marketplace.
- Medical diagnostics.
- Exact ball trajectory.

These would dilute the core proposition before it is proven.

---

# 49. Delivery phases

## Phase 0 - Feasibility Lab

Goal: prove or reject Watch-only contact-quality sensing.

Deliverables:

- Watch high-rate logger.
- Workout session.
- Manual trial labels.
- Raw export.
- Basic V40/impact metrics.
- GO/NO-GO report.

## Phase 1 - Measurement MVP

Only after GO.

Deliverables:

- Automatic impact detection.
- Initial Contact Quality score.
- Session storage.
- Basic Watch session UI.
- iPhone session viewer.
- Personal calibration.

## Phase 2 - Intelligent MVP

Deliverables:

- Personal baseline/trends.

- Apple Intelligence session summaries.
- Deterministic AI fallback.
- Explain This Session.
- Initial stroke-family classification if validated.

### Phase 3 - Premium Beta

Deliverables:

- Full premium iOS 27/watchOS 27 design.
- Ask Coach.
- Compare sessions.
- Widgets/complication as approved.
- Expanded physical validation.
- Battery and reliability optimization.
- Privacy/export controls.

### Phase 4 - Production V1

Deliverables:

- Validated scoring model.
- Stable Watch+iPhone sync.
- Production AI behavior/evaluations.
- Accessibility completion.
- App Store readiness.

### Phase 5 - Advanced roadmap

Candidates:

- Better personalized Core ML models.
- Coach mode.
- More precise stroke-family recognition.
- Multi-racket learning.
- Optional external-data integrations.

---

## 50. Definition of Done for a product feature

A feature is not Done because it compiles or looks finished.

For applicable features, Done requires:

- Requirement mapped to implementation.
- Unit/integration tests passing.
- No relevant full-suite failures.
- Exact committed code state known.
- Independent review complete where required.
- Physical Watch/iPhone evidence complete for physical behavior.
- Accessibility checked.
- Failure states tested.

- Documentation updated.
- No known blocker hidden behind a retry or fallback.

---

## 51. Acceptance criteria for the product concept

The product concept is successful if it can deliver all of the following without external racket hardware:

1. Reliably capture useful high-rate wrist impact data during padel.
2. Distinguish ball contact from non-contact swings with validated confidence.
3. Distinguish cleaner/central from poorer/off-centre contact with useful repeatability.
4. Produce a simple score that users understand and that is stable enough to track personal improvement.
5. Preserve uncertainty rather than fabricate precision.
6. Provide immediate Watch feedback without disrupting play.
7. Provide valuable post-session insight on iPhone.
8. Use Apple Intelligence to explain and coach from measured facts, not to invent them.
9. Work meaningfully without network/AI.
10. Present a premium, futuristic, native Apple design consistent with iOS 27/watchOS 27.

---

## 52. Product success metrics - later validation

Do not lock commercial targets before beta, but evaluate:

- Percentage of sessions completed without sensor failure.
- High-confidence stroke coverage.
- Classification accuracy by validated class.
- User agreement with clean/poor classification.
- Session-to-session repeatability.
- Battery consumption per hour.
- Sync reliability.
- AI factual-grounding rate.
- User return rate.
- Percentage of users who understand the score without explanation.

---

## 53. Open decisions

The following are intentionally not fixed yet:

- Final product name and brand.
- Final Contact Quality formula.
- Final scoring thresholds.
- Exact calibration protocol after feasibility.
- Whether 200 Hz device motion materially improves scoring.
- Final stroke classes.
- Final data-retention duration.
- Whether PCC entitlement will be available/used at launch.

- Commercial model/subscription structure.
- Whether users can share data with coaches in V1.

These shall be resolved through evidence and product decisions, not assumptions.

---

## 54. Implementation governance relationship

This Product/Design Bible defines **what** the product must be. It does not replace repository engineering governance, which defines **how** changes are authorized and evidenced.

Before each coding, test, review, Git, documentation or handover package:

- Read `AGENTS.md` if present.
- Read the repository-identified engineering governance document.
- Read the active Gate Card/equivalent project-control document.
- Reconcile branch, base, HEAD, upstream/remote HEAD, ahead/behind, worktree, changed files and current evidence.

Risk Level 3 applies to any app domain where failure can corrupt core state/evidence or undermine product trust, including:

- High-rate sensor/workout lifecycle.
- Stroke/scoring authority.
- Watch/iPhone sync and data integrity.
- Background/lifecycle/concurrency behavior.
- HealthKit permission/state handling.
- AI evidence-linkage if AI output could misstate measured results.
- Beta/readiness claims.

Required sequence for Risk Level 3 work:

**Authority → Risk map → Test contract → Implementation → Local automation → Clean CI → Independent review → Physical devices → PR/merge → Closure**

No Git state-changing action is implied by this document. Commit, push, PR, merge or branch deletion require the explicit approval process defined by repository governance.

---

## 55. Red-team completeness gate

This version has been reviewed against the failure modes most likely to make the product misleading, unbuildable, unreliable or non-premium.

### 55.1 Mandatory truths

The implementation must preserve all of the following:

- The Watch is worn on the racket hand.
- High-rate sensing is capability-checked at runtime.
- High-rate sensor delivery is batched; the app does not promise impossible instantaneous feedback.
- The Watch cannot observe the full laboratory vibration spectrum; it measures a band-limited wrist signal.
- A clean-contact score is an inference, not a direct measurement of sweet-spot position.

- Contact Quality is not court accuracy.
- Personal calibration and confidence are first-class concepts.
- AI never changes evidence or scores.
- A session must remain useful without AI or network.
- Physical device evidence is mandatory for sensor, workout, sync, battery, haptic and Watch usability claims.

## 55.2 Red-team blocking risks

These risks must be explicitly closed before production V1:

| Risk | Why it matters | Required closure |
|---|---|---|
| Wrist signal too weak/noisy | Core product would not be trustworthy | Phase 0 GO/NO-GO physical study |
| User self-label bias | Could make the model look better than it is | Controlled impact test + blinded/alternating labels where practical |
| Racket/player variability | Universal threshold may fail | Personal baseline + multi-user beta validation |
| 800 Hz bandwidth limit | Cannot see all racket modes | Use only validated low-band features; no full-spectrum claims |
| Batched delivery latency | Haptic/score may feel delayed | Physical UX timing test; design for post-shot, not during-shot feedback |
| Watch fit/strap variability | Alters transmission to sensor | Onboarding, calibration, beta matrix |
| Duplicate/lost sync data | Corrupts history/coaching | Idempotent IDs, acknowledgement protocol, reconnect tests |
| AI hallucination | Damages trust | Tool-grounded facts, structured output, evaluation suite, deterministic fallback |
| Beta API drift | Build may break before release | Feature flags + revalidation on shipping SDK |
| Health/privacy non-compliance | App Review/reputation risk | Privacy review, minimal HealthKit, no health-data advertising use |

## 55.3 Completeness decision

This document is sufficiently complete to guide architecture, UX, design, implementation packages, testing and staged delivery **provided that exact algorithm thresholds and final commercial decisions remain evidence-driven**. Those items are intentionally not invented here.

---

# 56. Product state machines

The app shall have explicit state machines rather than loosely coupled booleans.

## 56.1 Watch session state

Authoritative states:

```
idle -> preparing -> ready -> active -> paused -> ending -> completed
```

Failure/recovery states:

`permissionRequired`, `unsupported`, `sensorUnavailable`, `workoutFailed`, `interrupted`, `recoverableSession`, `syncPending`.

Rules:

- **Ready** is only reachable after workout + sensor capability + sensor stream + local persistence are verified.
- **Active** owns the authoritative live Session ID.
- **Paused** shall not create authoritative stroke events.
- **Ending** is idempotent; repeated user taps cannot create duplicate sessions.
- **Completed** requires local persistence, not phone sync.
- **SyncPending** may follow Completed and is not an error state.

## 56.2 Stroke-event state

```
candidateMotion -> candidateImpact -> validatedImpact -> featuresExtracted -> scored -> persisted -> syncPending -> synced
```

Alternative exits:

`rejectedNoise`, `airSwing`, `uncertain`, `corruptWindow`.

## 56.3 AI insight state

```
notRequested -> preparingFacts -> modelAvailableCheck -> generating -> validated -> displayed
```

Fallback exits:

`deterministicFallback`, `quotaUnavailable`, `offline`, `unsupported`, `generationFailed`.

AI state must never modify Session or Stroke authority.

---

# 57. Complete screen and UX catalogue

Every production screen shall have defined **loading, empty, normal, degraded and error** states where applicable.

## 57.1 iPhone onboarding

### Screen O1 - Welcome

Purpose: one-sentence value proposition.

Required:

- Premium hero visual.

- "Measure cleaner contact from your racket wrist" level of clarity.
- Continue action.

Do not show technical sensor language here.

### Screen O2 - Racket hand

Required:

- Right / Left choice.
- Visual Watch placement guidance.
- Confirmation that Watch must be on racket wrist.

### Screen O3 - Watch fit

Required:

- Snug sports fit guidance.
- Simple visual.
- No unsupported strap-tightness detection claim.

### Screen O4 - Permissions

Explain separately:

- Why HealthKit workout permission is required.
- Why Motion access is required.
- What the app does not read unless later approved.

Request permissions contextually rather than presenting a wall of permission dialogs on first launch.

### Screen O5 - Watch readiness

Display:

- Watch app installed/not installed.
- Supported high-rate sensor / unsupported.
- Required OS status.
- Clear repair action.

### Screen O6 - Racket profile

Optional in initial onboarding; skippable.

### Screen O7 - Calibration introduction

Explain that calibration helps compare the player against themselves.

### Screen O8 - Ready

Show one action: **Start first session**.

## 57.2 iPhone Today

Hero zone:

- Latest Contact Quality.
- Change versus comparable personal baseline.

- Session date/context.

Insight zone:

- Clean Contact Rate.
- Consistency.
- Relative Intensity relationship.

Coach zone:

- One Apple Intelligence insight or deterministic fallback.
- Source-data disclosure on tap.

Primary action:

- Start Session / Open Watch.

## 57.3 iPhone Sessions

Required states:

- No sessions.
- Synced sessions.
- Sync pending.
- Corrupt/import error.

List row:

- Date.
- Duration.
- Contact Quality.
- Stroke count.
- Confidence coverage.

## 57.4 iPhone Session Detail

Order:

1. Hero score and confidence coverage.
2. Short interpretation.
3. Quality-over-time chart.
4. Clean Contact Rate.
5. Quality by validated stroke family.
6. Quality by relative intensity.
7. AI Coach summary.
8. Stroke Explorer.
9. Export / delete.

## 57.5 iPhone Stroke Detail

Consumer mode:

- Contact Quality.
- Confidence.
- Stroke label if validated.
- Relative intensity.
- Personal-baseline delta.

- Explain This Stroke.

Lab mode additionally:

- Impact waveform.
- Raw sample metadata.
- V40 and approved features.
- Algorithm/model version.
- Sensor-rate diagnostics.

## 57.6 iPhone Coach

Required modules:

- Current focus.
- Improvement trend.
- Ask Coach.
- Suggested next-session focus.
- "Why this?" source facts.

The conversational UI must visibly separate user text from measured facts and generated advice.

## 57.7 iPhone Profile/Settings

Required:

- Racket hand.
- Rackets.
- Calibration.
- Watch readiness.
- Apple Intelligence availability.
- Haptic mode.
- Notifications.
- Data/export/delete.
- Privacy policy.
- How scoring works.
- Lab Mode access in development/beta only.

## 57.8 Watch Ready

Display only:

- Brand mark.
- Readiness state.
- Start.

A technical warning appears only when action is needed.

## 57.9 Watch Active

Primary visual:

- Large last Contact Quality or session state before first valid impact.
- Compact interpretation.
- Stroke count.

- Session clean-contact trend/percentage.

A score shall not flash for low-confidence strokes as if certain.

### 57.10 Watch Pause/End

Use large standard controls and an explicit End confirmation that does not risk accidental session loss.

### 57.11 Watch Summary

Required:

- Contact Quality.
- Clean Contact Rate.
- Stroke count.
- One improvement observation.
- Sync state if pending.

---

## 58. Premium design specification

### 58.1 Signature visual language - Impact Field

The brand should have a recognizable visual motif derived from the physics of contact rather than generic sports decoration.

**Impact Field** is a restrained concentric/elliptical energy pattern used around key score moments and hero surfaces. It should resemble precision measurement, not a neon explosion.

Rules:

- Generated from abstracted geometry, never presented as literal sensor data unless explicitly labelled.
- Subtle depth, low-opacity spectral gradient and precise motion.
- No continuous decorative animation during active play.
- Reduced Motion replaces animated propagation with a static state change.

### 58.2 Color tokens

Semantic tokens - exact RGB values remain design-system implementation details and must pass contrast testing:

| Token | Purpose |
|---|---|
| Surface/Obsidian | Primary dark content background |
| Surface/Elevated | Secondary content surface |
| Text/Primary | High-priority labels/numbers |
| Text/Secondary | Context labels |
| Accent/Ion | Performance/measurement emphasis |
| Accent/Violet | Intelligence/Coach |
| State/Clean | Positive contact state |
| State/Uncertain | Caution/low confidence |
| State/Attention | Poor-contact attention state |

No consumer meaning may rely on hue alone.

## 58.3 Spacing and composition

Use an 8-point conceptual rhythm where practical, while allowing native Apple components to control their own metrics.

Preferred content rhythm:

- 8 pt micro spacing.
- 16 pt component spacing.
- 24/32 pt section breathing room.

Do not hard-code system navigation/control dimensions.

## 58.4 Typography hierarchy

- Watch hero score: largest practical system numeric style for device size.
- iPhone hero score: large Display/Title treatment with monospaced digits.
- Interpretation: short semibold label.
- Technical units: secondary, never competing with the score.
- Charts: legible axis labels, minimal grid noise.

## 58.5 Liquid Glass rules

Liquid Glass belongs mainly to the functional/navigation layer.

Use for:

- Tab bar.
- Navigation controls.
- Primary floating action.
- Menus/popovers.
- Interactive sliders/toggles when system behavior applies.

Do not use as the background of every metric card. Standard content materials and negative space carry the premium visual hierarchy.

## 58.6 Motion timing targets

Motion must feel immediate, not theatrical.

Initial design targets to validate on device:

- Micro state transitions: approximately 120-220 ms.
- Hero score transition: approximately 220-450 ms.
- Post-impact visual settle: less than 700 ms after the score is available.

These are design targets, not sensor-latency claims.

## 58.7 Haptic system

Three settings:

- Off.
- Important Only - default.
- Coach Mode.

Important Only:

- Session start/end.
- Sensor failure requiring attention.
- Optional milestone.

Coach Mode may add a subtle poor-contact pattern, but only after physical testing confirms it does not distract from play.

### 58.8 Premium anti-patterns

Release blocker for design review if the UI becomes:

- A wall of cards.
- Dashboard-first.
- Over-glassed.
- Excessively neon.
- Data-dense on Watch.
- Animated during every sensor event.
- Inconsistent with Dynamic Type / Reduce Motion / Increase Contrast.

---

## 59. Engineering blueprint and recommended repository structure

The final repository should separate product authority, shared domain logic, Watch-specific sensing and iPhone-specific presentation.

Recommended conceptual structure:

```
PadelPerformance/
  AGENTS.md
  docs/
    PRODUCT_REQUIREMENTS_AND_DESIGN_BIBLE.pdf-or-md
    engineering-governance.*
    gate-cards/
  App/
    iOS/
    watchOS/
  Packages/
    Domain/
    MotionAnalysis/
    Scoring/
    Persistence/
    Sync/
    Intelligence/
    DesignSystem/
  Tests/
    Unit/
    Integration/
    Fixtures/
  Tools/
    ExportAnalysis/
    Validation/
```

Actual repo layout is established by the repository and approved architecture; this is a recommended starting shape, not permission to refactor an existing repo.

### 59.1 Targets

Expected Xcode targets/schemes:

- iPhone app target.
- Watch app target.
- Shared unit-test targets/packages.
- Optional UI test target.
- Development-only Lab/diagnostic configuration through feature flags rather than a duplicated product.

### 59.2 Swift engineering rules

- Swift strict concurrency enabled when supported by the chosen toolchain.
- Prefer `async/await` and actors for isolated mutable state.
- No blocking high-rate work on `MainActor`.
- No force unwraps in authoritative production paths unless impossible state is proven and documented.
- Dependency boundaries use protocols where they materially improve testing.
- Apple-first dependency strategy; third-party SDKs require explicit privacy/security/product justification.
- `OSLog` categories for lifecycle, sensor, scoring, sync and intelligence with privacy-safe fields.
- Raw sensor samples must never be dumped into production logs.

### 59.3 Build configurations

Minimum:

- Debug.
- Beta/TestFlight.
- Release.

Debug may expose Lab Mode and verbose diagnostics. Beta may expose controlled evidence export. Release must disable internal-only controls and raw logging not intended for consumers.

---

## 60. Permissions, capabilities and entitlements

### 60.1 Watch/iPhone capabilities

Expected capabilities subject to actual feature scope:

- HealthKit.
- App Groups only if a validated cross-process need exists.
- Background modes only when Apple platform rules and actual feature behavior justify them.
- Private Cloud Compute entitlement only if approved and available.
- App Intents/WidgetKit as required by shipped system integrations.

Do not enable capabilities "just in case."

### 60.2 Required usage descriptions

Before implementation review, the project must contain clear user-facing reasons for any requested capability, including as applicable:

- Motion usage.
- Health data read/write usage.

Exact text must match actual data use and be reviewed before App Store submission.

### 60.3 Permission timing

Request a permission immediately before the user needs the feature and after a plain-language pre-permission explanation where appropriate.

Do not:

- Ask for unrelated health categories.
- Ask for permissions before explaining the benefit.
- Treat a denied read permission as a technical crash condition.

---

## 61. Sensor and signal-processing implementation contract

### 61.1 Raw accelerometer record

Each retained high-rate sample shall include at minimum:

- Monotonic sensor timestamp.
- X, Y, Z acceleration as delivered by the API.
- Batch identifier / capture sequence metadata sufficient to detect gaps.
- Session identifier.

### 61.2 Resultant acceleration

Where orientation independence is helpful, compute:

```
aMag = sqrt(x^2 + y^2 + z^2)
```

Do not assume this replaces axis-specific information; both representations may be useful.

### 61.3 Filtering

The feasibility implementation may compare:

- Raw axis/resultant windows.
- High-pass or band-pass features that isolate impact transients.

Every filter must be:

- Versioned.
- Deterministic.
- Tested on synthetic fixtures.
- Applied to the production data path, not a copied test implementation.

### 61.4 Impact timestamp

Initial impact detection may use the strongest validated filtered acceleration peak within a candidate-swing window.

It must include:

- Minimum peak criteria.
- Refractory/debounce period.
- Plausible swing-window checks.
- Corrupt/gap rejection.

Thresholds remain empirical until physical data exists.

### 61.5 V40 feature

For a validated impact timestamp `t0`, calculate a provisional vibration-energy feature over the first 40 ms after contact using the chosen acceleration representation:

```
V40 = integral(a(t)^2 dt), t0 ... t0 + 0.040 s
```

Implementation shall use actual sample timestamps rather than assume perfect spacing when practical.

### 61.6 Additional candidate features

Only after raw evidence review:

- Peak acceleration.
- RMS acceleration in defined windows.
- Early/late energy ratio.
- Decay envelope.
- Axis-energy ratios.
- Pre-impact motion energy.
- Device-motion angular velocity.
- Rotation along gravity.
- Repeatability distance to personal baseline.

### 61.7 Spectral features

Any spectral feature must respect the 800 Hz sample rate and cannot interpret frequencies above 400 Hz. Features near the Nyquist limit require special caution and physical validation.

### 61.8 Sample-gap handling

A stroke window containing a material sample gap shall be marked invalid/uncertain rather than silently interpolated into an authoritative score.

---

## 62. Feasibility logger and physical study protocol

Phase 0 exists to answer **one question** before consumer development:

> *Can the Apple Watch on the racket wrist separate useful contact-quality patterns strongly enough to justify the product?*

### 62.1 Logger requirements

The Watch feasibility logger shall provide:

- Sensor capability status.
- Active workout state.
- Actual samples received per batch/second.
- Manual trial label.
- Session/trial ID.
- Raw impact window retention.

- Derived provisional features.
- CSV/JSON export path.
- No consumer-facing AI dependency.

Manual labels:

- Air Swing.
- Centre Impact.
- Off-Centre Impact.
- Clean Real Stroke.
- Poor Real Stroke.

## 62.2 Test setup

Keep constant where practical:

- Same Apple Watch.
- Same wrist.
- Same strap and snugness.
- Same racket.
- Same ball.
- Same racket grip.
- Similar environment.

Record deviations in trial notes.

## 62.3 Test A - sensor acquisition

Perform a short active workout capture.

PASS requires:

- Supported high-rate stream.
- Stable batch delivery.
- Plausible sample counts/timestamps.
- Persisted raw data.
- No unexplained material gaps.

## 62.4 Test B - air swing versus impact

Minimum simple run:

- 5 air swings.
- 5 obvious ball impacts.

PASS requires all obvious impacts to produce a separable impact signature and no air swing to be misrepresented as a high-confidence impact in this basic run.

If this fails, do not progress to scoring.

## 62.5 Test C - controlled centre versus off-centre

Use a consistent self-drop or controlled tap method.

Recommended first dataset:

- 15 centre impacts.

- 15 off-centre impacts, inside the playable face and away from direct frame strikes.

Prefer alternating/randomized order rather than doing all centre trials first if practical.

Analysis:

- Compare distributions, not only mean values.
- Inspect raw windows for obvious artifacts.
- Use a simple threshold/classifier defined without peeking at every test label.

Initial GO threshold:

- At least 80% correct separation on the defined held-out/simple validation rule.

### 62.6 Test D - real strokes

After controlled PASS:

- 10 deliberately clean self-fed/simple strokes.
- 10 deliberately poor/off-centre strokes.

The human label is less objective than the controlled test, so this result supports but does not replace Test C.

PASS target:

- At least 80% correct classification with no systematic capture failure.

### 62.7 Borderline rule

To preserve a final binary decision without pretending a tiny sample is statistically perfect:

- *=80% on controlled and real tests → **GO**.*
- <70% on either after capture problems are excluded → **NO-GO**.
- 70-79% → perform one predefined repeat set only; final combined result decides GO/NO-GO.

No repeated retries until a passing result appears.

### 62.8 Final Phase 0 decision

GO permits Phase 1 development.

NO-GO stops the Watch-only contact-quality product direction unless a materially different sensing architecture is explicitly approved.

---

## 63. Apple Intelligence implementation guide

Apple Intelligence is included because it can turn trustworthy sports measurements into understandable, personal coaching. It is **not** included as a marketing badge.

### 63.1 Highest-value use cases

#### A. Session Coach - highest priority

Input facts:

- Session Contact Quality.

- Clean Contact Rate.
- Confidence coverage.
- Stroke-family changes.
- Quality-by-intensity changes.
- Personal baseline deltas.

Output:

- 2-4 sentence plain-language summary.
- One evidence-based focus for next session.

User value: the player immediately understands what matters without interpreting charts.

### B. Explain This Stroke

Input facts:

- Stroke score.
- Confidence.
- Impact features.
- Baseline delta.
- Stroke family/intensity if validated.

Output:

- Concise explanation of which measured patterns drove the result.

User value: converts an abstract score into learning.

### C. Ask Coach

The user can ask natural-language questions about their own data.

Examples:

- "Why was today worse?"
- "Am I getting more consistent?"
- "What happens when I hit harder?"
- "Which stroke should I work on next?"

The model must call approved data tools to answer.

### D. Weekly Progress Narrative

Summarize meaningful change across sessions only when enough comparable data exists.

### E. Glanceable Watch summary

At session end, optionally provide one short sentence. Do not run a long conversation on the wrist.

## 63.2 Model-routing policy

Default order:

1. Deterministic analytics first.
2. `SystemLanguageModel` when available and evaluation proves sufficient.
3. PCC only when stronger reasoning/context is genuinely needed and entitlement/eligibility/quota/network allow.

Do not send every raw stroke or raw 800 Hz sample into a language model.

## 63.3 AI data contract

The model receives structured facts, for example:

```
{
  "sessionID": "...",
  "contactQuality": 81,
  "cleanContactRate": 0.78,
  "confidenceCoverage": 0.92,
  "forehandDeltaVsBaseline": 0.09,
  "highIntensityDeltaVsBaseline": -0.13,
  "knownLimitations": ["ball placement unavailable"]
}
```

The model does not receive the authority to write these values back.

## 63.4 Approved tool contracts

Tools shall return typed, bounded data and shall never accept arbitrary database queries from model text.

Recommended tools:

- `getSessionSummary(sessionID)`
- `compareSessions(sessionA, sessionB)`
- `getTrend(metric, dateRange)`
- `getStrokeStats(strokeFamily, dateRange)`
- `getQualityByIntensity(dateRange)`
- `getPersonalBaseline(strokeFamily)`
- `getEvidenceForInsight(insightID)`

Tool result rules:

- Include confidence/coverage.
- Include algorithm/model versions when relevant.
- Reject unsupported metrics.
- Avoid returning raw HealthKit data unless the feature has explicit need and approval.

## 63.5 Prompt/instruction rules

System instructions shall require the model to:

- Use only supplied/tool facts for claims about the user.
- State when data is insufficient.
- Avoid medical diagnosis.
- Avoid claiming ball placement, spin or exact racket speed.
- Give at most one primary training focus unless the user asks for detail.
- Use clear, non-patronizing language.
- Distinguish observation from recommendation.

## 63.6 Structured output

User-facing AI cards should use typed/generated structures, for example:

```
CoachInsight
- headline
- evidenceSummary
- nextFocus
```

```
- confidenceNote
- sourceMetricIDs
```

Do not parse free-form prose to decide application state.

## 63.7 Context budget

On-device context is limited. The app shall summarize historical data into compact analytical facts before prompting.

PCC may support larger context and reasoning, but use is still bounded by entitlement and per-user quota.

## 63.8 Availability states

UI must explicitly handle:

- Apple Intelligence disabled.
- Device not eligible.
- Model not ready.
- Unsupported language/locale.
- Network unavailable for PCC.
- PCC quota approaching/exceeded.
- Entitlement unavailable.
- Generation error.

Every state has a deterministic fallback.

## 63.9 AI evaluation suite

Maintain fixed and generated evaluation cases covering:

- Metric fidelity.
- No invented stats.
- Correct tool selection.
- Low-data refusal/uncertainty.
- Contradictory input handling.
- Unsupported biomechanics rejection.
- Medical-question boundary.
- Concision on Watch.
- Locale behavior.
- PCC/on-device consistency where both are allowed.

Prompt/model changes invalidate affected AI evidence and must be re-evaluated.

## 63.10 User transparency

The app should provide a simple explanation:

> *"Coach insights are generated from your measured session statistics. AI does not change your scores."*

That statement is part of product trust.

---

# 64. Siri, App Intents and system intelligence

App Intents make core product actions discoverable to Siri, Spotlight, Shortcuts, widgets and other system experiences.

Initial intents after the core product is stable:

- Start Padel Session.
- Show Latest Session.
- Show Current Contact Quality Trend.
- Compare Last Two Sessions.
- Show Current Training Focus.

Rules:

- Intents call the same authoritative domain services as the app UI.
- "Start Session" shall not bypass readiness checks.
- Read intents expose only data appropriate for the current privacy context.
- Do not create broad write actions that could alter or delete session evidence through ambiguous natural language.

Future Ultra Action Button integration may use an App Intent if it improves real court use and is physically validated.

---

# 65. Security, privacy and compliance blueprint

## 65.1 Data classes

Classify at minimum:

- Raw motion snippets - sensitive performance/behavioral data.
- Derived stroke metrics - personal performance data.
- HealthKit workout metadata - health/fitness data.
- AI insights - derived personal performance interpretation.
- Diagnostics - technical data, privacy-filtered.

## 65.2 Local protection

Use platform data protection for persisted personal data. Secrets/tokens, if introduced later, belong in Keychain and must never be hard-coded.

## 65.3 Network

If any network/backend is introduced:

- TLS only.
- No raw personal sensor data in query strings/logs.
- Authentication/authorization reviewed separately.
- Retention and deletion defined before launch.

### 65.4 Logging

Production logs may contain:

- Stable internal event code.
- Session lifecycle state.
- Error category.
- Count/rate summaries that cannot reconstruct a user's raw stroke data.

Production logs shall not contain:

- Raw sensor arrays.
- Full AI prompts with personal statistics unless explicitly protected and approved for diagnostics.
- HealthKit sample values unrelated to a logged error.

### 65.5 Analytics

Product analytics is optional and must be privacy-minimal.

Never use HealthKit or detailed motion data for advertising or behavioral targeting.

Prefer coarse product events such as:

- Session started/completed.
- AI insight viewed.
- Calibration completed.

No analytics SDK may access HealthKit/raw sensor data unless explicitly approved and compliant.

### 65.6 Deletion and export

Deleting a session shall delete its app-owned stroke data, raw retained windows and AI insights according to the documented retention model.

Export must clearly distinguish:

- Measured data.
- Derived scores.
- AI-generated text.

---

## 66. Persistence, sync and migration blueprint

### 66.1 Identifiers

Create UUIDs on the authoritative device before sync for:

- Session.
- Stroke.
- Calibration.
- AI insight.

### 66.2 Watch sync protocol

At minimum:

- Watch persists first.
- Transfer carries stable IDs + version/schema metadata.
- iPhone upserts idempotently.
- iPhone acknowledges durable receipt.
- Watch retains pending records until acknowledgement or defined retention expiry.
- Reconnect/retry never creates a second stroke/session with a new ID.

### 66.3 Conflict rule

The phone must not silently overwrite an authoritative Watch stroke measurement with an older copy.

Derived iPhone analytics may be recomputed, but raw/authoritative stroke facts follow the versioned source-of-truth rules.

### 66.4 Schema migrations

Every persistent schema change requires:

- Migration test from every supported prior production schema.
- Roll-forward strategy.
- Failure behavior that preserves user data.
- Backup/export consideration for beta research data.

### 66.5 Historical algorithm versions

Historical sessions retain the algorithm/model version used to create the score.

If the product introduces score normalization across versions, the transformation must be explicit and auditable rather than silently rewriting history.

---

## 67. Non-functional requirements and performance budgets

Targets remain subject to physical evidence; they are engineering goals, not marketing promises.

### 67.1 Reliability

- No completed session lost after successful local persistence.
- No duplicate authoritative strokes after reconnect.
- App crash during sync does not corrupt already-persisted Watch evidence.
- Interrupted workout has a recoverable or explicitly abandoned state.

### 67.2 Latency

Because high-rate data arrives in roughly one-second batches, design for post-shot feedback rather than during-contact intervention.

Initial physical target:

- Score/label appears within **1.5 seconds of impact in normal conditions**, subject to API batch timing.

If the measured platform behavior cannot meet this, the product shall adjust UX rather than hide the latency.

### 67.3 Watch UI responsiveness

- Main-thread interactions remain smooth during capture.
- High-rate buffers never drive per-sample UI updates.
- Controls respond immediately to touch independent of analysis work.

### 67.4 Battery

Beta shall measure:

- Percentage battery consumed per 30/60/90-minute session.
- Thermal behavior.
- Effect of Always On display state.
- Effect of device-motion stream and AI features.

No battery claim is approved until a representative physical matrix is complete.

### 67.5 Storage

Raw data retention must be bounded. The app shall calculate worst-case storage before enabling a capture mode in Beta/Release.

### 67.6 Accessibility/performance interaction

Reduced Motion/Transparency settings must not degrade semantic clarity or block the primary action.

---

## 68. Test, evidence and release matrix

### 68.1 Automated unit tests

Mandatory domains:

- Timestamp math.
- Magnitude calculation.
- Filter fixtures.
- Impact-window extraction.
- V40 calculation.
- Gap rejection.
- Confidence thresholds.
- Session aggregation.
- Persistence round-trip.
- Sync idempotency.
- Schema migration.
- AI tool data contracts.
- Deterministic AI fallback.

### 68.2 Integration tests

Mandatory:

- Workout lifecycle.

- Sensor lifecycle.
- Pause/resume/end.
- Interrupted session recovery.
- Watch persistence before sync.
- Reconnect and duplicate delivery.
- AI availability routing.
- PCC quota fallback simulation when shipped.

## 68.3 UI tests

Focus on stable critical flows rather than brittle visual snapshots:

- Onboarding.
- Permission explanation.
- Start/stop.
- Session summary.
- Delete/export.
- AI unavailable state.

## 68.4 Accessibility tests

- VoiceOver labels/order.
- Dynamic Type.
- Reduce Motion.
- Reduce Transparency.
- Increase Contrast.
- Color differentiation.
- Large touch targets.

## 68.5 Physical device gates

Required for relevant releases:

- Apple Watch Ultra 2 + paired iPhone baseline.
- Racket wrist physical capture.
- Workout/background/lifecycle behavior.
- Haptics.
- Always On usability.
- Disconnect/reconnect.
- Battery.
- Sensor rate/sample gaps.

Additional Watch/iPhone models are added before broad beta/public claims.

## 68.6 AI gates

- Fixed evaluation suite passes.
- No blocking hallucination on defined cases.
- Tool-call failures degrade safely.
- On-device/PCC availability states tested.
- Prompt/model versions recorded.

### 68.7 Release evidence rule

Evidence is valid only for the exact code state tested. Production, test-seam, model, prompt, entitlement, build-setting or migration changes invalidate the affected evidence.

---

## 69. A-to-Z delivery plan

This is the recommended product-development sequence. Exact repo governance controls execution.

### Phase A - Project authority and blank app

Deliver:

- Clean Xcode project.
- Git repository on a non-main working branch after authority is established.
- `AGENTS.md` if the repo requires it.
- Engineering governance document.
- Active Gate Card.
- This Product/Design Bible stored under `docs/`.

Gate:

- Clean known branch/HEAD/worktree and approved Phase 0 scope.

### Phase B - Feasibility logger

Deliver:

- HealthKit workout lifecycle.
- Motion authorization.
- High-rate `CMBatchedSensorManager` capture.
- Lab UI.
- Manual labels.
- Raw export.
- Sample-rate/gap diagnostics.
- Automated math tests.

Gate:

- Clean automated build/tests + independent review before physical evidence.

### Phase C - Physical GO/NO-GO

Execute Section 62 exactly.

Gate:

- GO → Phase D.
- NO-GO → stop consumer build and document result.

### Phase D - Measurement MVP

Deliver:

- Automatic impact detector.
- Contact Quality v1.
- Confidence.
- Calibration v1.
- Session persistence.
- Basic consumer Watch UI.
- Basic iPhone history.

Gate:

- Multi-session physical stability and scoring repeatability.

## Phase E - Sync and history

Deliver:

- Watch/iPhone idempotent synchronization.
- Session repository.
- Trends/charts.
- Versioning/migrations.

Gate:

- Disconnect/reconnect and duplicate-delivery tests + physical proof.

## Phase F - Premium design system

Deliver:

- Final navigation architecture.
- Impact Field visual identity.
- Liquid Glass adoption.
- Motion/haptics.
- Accessibility.
- Icon/brand.

Gate:

- Design review on physical Watch/iPhone in all required accessibility appearances.

## Phase G - Apple Intelligence

Deliver in order:

1. Deterministic session narrative.
2. On-device Session Coach.
3. Explain This Stroke/Session.
4. Ask Coach tools.
5. PCC only if evaluation proves a meaningful benefit and entitlement is available.
6. App Intents/Siri integrations.

Gate:

- AI evaluation + privacy + availability/fallback proof.

### Phase H - Beta

Deliver:

- TestFlight package.
- Lab evidence export for selected testers.
- Multi-user/racket/band validation.
- Battery matrix.
- Crash/reliability review.
- Marketing-claim truth check.

Gate:

- No full-suite failures, independent review of exact HEAD, relevant physical gates complete.

### Phase I - Production V1

Deliver:

- Final supported-device matrix.
- Final scoring/versioning policy.
- Final privacy policy/App Privacy data.
- Final App Store assets/review notes.
- Final shipping-SDK validation.

Gate:

- App Store submission approval decision only after all mandatory evidence is current.

---

## 70. Requirements traceability and acceptance language

Every mandatory statement in this document is an acceptance criterion unless explicitly labelled as a candidate, example, future option or open decision.

During development, each scoped requirement shall be marked:

- **VERIFIED** - evidence exists for exact code state.
- **CONTRADICTED** - evidence shows it is not met.
- **UNKNOWN** - not yet proven.
- **BLOCKED** - cannot progress due to a known dependency/failure.

A feature cannot be called "ready", "safe", "validated", "premium-complete" or equivalent while any mandatory requirement in its scope remains CONTRADICTED or BLOCKED.

Design acceptance must include both:

- Static visual review.
- Physical interactive review on the intended devices.

Algorithm acceptance must include both:

- Automated deterministic tests.
- Relevant real-world sensor evidence.

AI acceptance must include both:

- Evaluation results.

- Grounded-data inspection for representative real sessions.

---

## 71. Reference basis

The requirements above are informed by the following authoritative or research sources:

- Apple Developer - WWDC23: What's new in Core Motion.
- Apple Developer - `CMBatchedSensorManager` documentation.
- Apple Developer - HealthKit `HKWorkoutSession` documentation.
- Apple Developer - What's new in iOS 27.
- Apple Developer - What's new in watchOS 27.
- Apple Developer - Foundation Models framework documentation.
- Apple Developer - Adding server-side intelligence with Private Cloud Compute.
- Apple Developer - Liquid Glass technology overview and Human Interface Guidelines.
- Apple Developer - Designing for watchOS.
- Apple Developer - Charts, Motion, Materials, SF Symbols and App Icon guidance.
- Sensors (2023) - "Identifying the Sweet Spot of Padel Rackets with a Robot."
- PubMed - "Transfer of tennis racket vibrations onto the human forearm."

---

Current truth-check references (24 Aug 2026):

- Apple Developer, "What's new in Core Motion" (WWDC23): https://developer.apple.com/videos/play/wwdc2023/10179/
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## 72. Final product statement

The app shall be designed and engineered as a **premium Apple-native padel performance instrument**.

Its core value is not "AI" and not "vibration graphs." Its core value is:

> ***Turn the motion already available on a player's racket wrist into trustworthy, understandable feedback that helps the player make cleaner, more repeatable contact.***

The Apple Watch measures.
The scoring engine interprets.
The iPhone reveals patterns.
Apple Intelligence explains and coaches.
The player improves.

That sequence is the product.